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Loss distributions

Syllabus objectives

- 1.1 Loss distributions, with and without risk sharing
 - 1.1.1 Describe the properties of the statistical distributions which are suitable for modelling individual and aggregate losses.
 - 1.1.5 Estimate the parameters of a failure time or loss distribution when the data is complete, or when it is incomplete, using maximum likelihood and the method of moments.
 - 1.1.6 Use R to fit a statistical distribution to a dataset and calculate appropriate goodness-of-fit measures.

0 Introduction

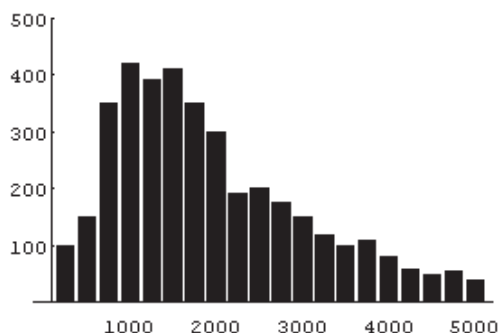
General insurance companies need to investigate claims experience and apply mathematical techniques for many purposes. These include:

- premium rating (*ie* deciding what premium rates to charge policyholders)
- reserving (*ie* assessing how much money should be set aside to cover the cost of claims)
- reviewing reinsurance arrangements
- testing for solvency (*ie* assessing the company's financial position).

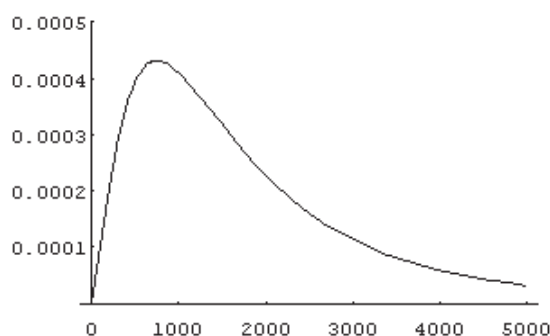
In this chapter we will look at loss distributions. These are statistical distributions that are used to model individual claim amounts. We will introduce some new distributions and we see how these can be fitted to observed claims data. We can then test for goodness of fit, and use the fitted loss distributions to estimate probabilities.

The total amount of claims in a particular time period is a quantity of fundamental importance to the proper management of an insurance company. The key assumption in all the models studied here is that the occurrence of a claim and the amount of a claim can be studied separately. Thus, a claim occurs according to some simple model for events occurring in time, then the amount of the claim is chosen from a distribution describing the claim amount.

Note carefully the distinction here. The frequency of claim amounts when plotted against size might look like this:



The statistical distributions in this chapter are used to approximate this distribution, which is called a *loss distribution*. For example, we might decide to use a loss distribution like this as an approximation to the claims arising in the graph above:



A range of statistical techniques can be used to describe the distribution of random variables. The object is to describe the variation in claim amounts by finding a loss distribution that adequately describes the claims that actually occur. As usual this can be done at two levels.

At a first level, it can be assumed that the claims arise as realisations from a known distribution. For example, it may be possible to assume that the logarithm of the claim amount follows, to a reasonable approximation, a normal distribution with known mean and known standard deviation. Knowledge of the claim amount process would be complete, and interest would then centre on the consequences for insurance. For example, claims above a certain level might trigger some reinsurance arrangements or claims below a certain level might never be lodged if a policy excess was in force.

A policy excess means that the policyholder has to pay the first part of any claim. For example, with car insurance in the UK the policyholder often has to pay the first £200 of any claim. The insurer pays the rest.

In practice the exact claims distribution will hardly ever be known. At this second level a standard method of proceeding is to assume that the claims distribution is a member of a certain family. The parameters of the family must now be estimated using the claim amount records by an appropriate method such as maximum likelihood. Complications will arise if large claims have been limited (reinsurance) or some small claims have not been lodged (policy excess).

We will consider the effects of reinsurance and policy excesses in [Chapter 18](#).

Many studies have been made of the kind of distribution that can be used to describe the variation in claim amounts.

The typical pattern is as shown in the histogram above, with a few small claims, rising to a peak, then tailing off gradually with a few very large claims.

The general conclusion is that claims distributions tend to be positively skewed and long tailed.

1 Simple loss distributions

In this section we will review some of the properties of the statistical distributions that are used to model claim amounts. In most cases we use a positively skewed, continuous distribution.

Recall that, for a continuous random variable, X :

- the cumulative distribution function (CDF) is:

$$F_X(x) = P(X \leq x)$$

- the probability density function (PDF) is:

$$f_X(x) = F'_X(x)$$

- probabilities can be expressed in terms of the PDF or CDF:

$$P(a \leq X \leq b) = \int_a^b f_X(x) dx = F_X(b) - F_X(a)$$

- the moment generating function (MGF) is:

$$M_X(t) = E(e^{tX}) = \int_x e^{tx} f_X(x) dx$$

The CDF, PDF and MGF may also be denoted without the subscript as $F(x)$, $f(x)$ and $M(t)$, respectively, provided the meaning is clear.

The formulae for the densities, the moments and the moment generating functions (where they exist) for the distributions discussed in this chapter are given in the *Formulae and Tables for Actuarial Examinations*.

In the *Tables*, the abbreviation DF is used for cumulative distribution function.

1.1 The exponential distribution

A random variable X has the exponential distribution with parameter $\lambda > 0$ if it has CDF:

$$F(x) = 1 - e^{-\lambda x}, \quad x > 0$$

In that case we write $X \sim \text{Exp}(\lambda)$.

The PDF is:

$$f(x) = \lambda e^{-\lambda x}, \quad x > 0$$

The mean and variance are $\frac{1}{\lambda}$ and $\frac{1}{\lambda^2}$ respectively.

The PDF can also be written as:

$$f(x) = \frac{1}{\mu} e^{-x/\mu}$$

where μ is the mean.

The MGF is:

$$M(t) = \left(1 - \frac{t}{\lambda}\right)^{-1}, \quad t < \lambda$$

All of these formulae are given on page 11 of the *Tables*.



Question

A portfolio of insurance policies contains two types of risk. Type I risks make up 70% of claims and give rise to loss amounts that are exponentially distributed with mean 500. Type II risks give rise to loss amounts that are exponentially distributed with mean 1,000.

Let X denote the amount of a randomly chosen loss. Determine $E(X)$, $\text{var}(X)$ and $M_X(t)$.

Solution

Since the amount of a loss depends on the type of risk from which it arises, we calculate $E(X)$ using the conditional expectation formula (from Subject CS1). This formula is given on page 16 of the *Tables*. In this case:

$$\begin{aligned} E(X) &= E(E(X | \text{Type})) = E(X | \text{Type I})P(\text{Type I}) + E(X | \text{Type II})P(\text{Type II}) \\ &= 500 \times 0.7 + 1,000 \times 0.3 = 650 \end{aligned}$$

Similarly:

$$E(X^2) = E(E(X^2 | \text{Type})) = E(X^2 | \text{Type I})P(\text{Type I}) + E(X^2 | \text{Type II})P(\text{Type II})$$

Now:

$$E(X^2 | \text{Type I}) = \text{var}(X | \text{Type I}) + [E(X | \text{Type I})]^2 = 500^2 + 500^2 = 500,000$$

Here we are using the fact that the variance of an exponential random variable is the square of its mean. So the variance of losses from Type I risks is 500^2 . We can use the same approach for Type II risks:

$$E(X^2 | \text{Type II}) = \text{var}(X | \text{Type II}) + [E(X | \text{Type II})]^2 = 1,000^2 + 1,000^2 = 2,000,000$$

So:

$$E(X^2) = 500,000 \times 0.7 + 2,000,000 \times 0.3 = 950,000$$

Hence:

$$\text{var}(X) = 950,000 - 650^2 = 527,500$$

Alternatively, we could calculate $\text{var}(X)$ using the conditional variance formula, which is also given on page 16 of the *Tables*:

$$\text{var}(X) = E[\text{var}(X | \text{Type})] + \text{var}[E(X | \text{Type})]$$

For notational convenience, let $V = \text{var}(X | \text{Type})$ and let $W = E(X | \text{Type})$. Then V has the following distribution:

v	500^2	$1,000^2$
$P(V = v)$	0.7	0.3

So:

$$E(V) = 500^2 \times 0.7 + 1,000^2 \times 0.3 = 475,000$$

In addition, we can calculate $\text{var}(W)$ from the distribution of W :

w	500	1,000
$P(W = w)$	0.7	0.3

We have:

$$E(W) = 500 \times 0.7 + 1,000 \times 0.3 = 650$$

$$E(W^2) = 500^2 \times 0.7 + 1,000^2 \times 0.3 = 475,000$$

and hence:

$$\text{var}(W) = 475,000 - 650^2 = 52,500$$

So:

$$\text{var}(X) = E[V] + \text{var}[W] = 475,000 + 52,500 = 527,500$$

Finally, we will consider the moment generating function of X . Again, we will use the conditional expectation formula:

$$\begin{aligned} M_X(t) &= E(e^{tX}) = E[E(e^{tX} | \text{Type})] \\ &= E(e^{tX} | \text{Type I})P(\text{Type I}) + E(e^{tX} | \text{Type II})P(\text{Type II}) \end{aligned}$$

$E(e^{tX} \mid \text{Type I})$ is the MGF of the exponential distribution with mean 500, ie:

$$E(e^{tX} \mid \text{Type I}) = (1 - 500t)^{-1}$$

Similarly, $E(e^{tX} \mid \text{Type II})$ is the MGF of the exponential distribution with mean 1,000, ie:

$$E(e^{tX} \mid \text{Type II}) = (1 - 1,000t)^{-1}$$

So:

$$M_X(t) = 0.7(1 - 500t)^{-1} + 0.3(1 - 1,000t)^{-1}$$

We can use R to simulate values from statistical distributions, plot their PDFs, and calculate probabilities and percentiles. An example involving the exponential distribution is given below.



Suppose we have an exponential distribution with parameter $\lambda = 0.5$. The R code for simulating 100 values is given by:

```
rexp(100, rate=0.5)
```

The PDF is obtained by `dexp(x, rate=0.5)` and is useful for graphing. For example:

```
plot(seq(0:5000), dexp(seq(0:5000), rate=0.5), type="l")
```

To calculate probabilities for a continuous distribution we use the CDF which is obtained by `pexp`. For example, to calculate $P(X \leq 2) = 0.6321206$ we use the R code:

```
pexp(2, rate=0.5)
```

Similarly, the quantiles can be calculated with `qexp`.

This code can be adapted to deal with other statistical distributions.

1.2 The gamma distribution

The random variable X has a gamma distribution with parameters $\alpha > 0$ and $\lambda > 0$ if it has PDF:

$$f(x) = \frac{\lambda^\alpha}{\Gamma(\alpha)} x^{\alpha-1} \exp(-\lambda x), \quad x > 0$$

In that case we write $X \sim \text{Ga}(\alpha, \lambda)$.

This may also be written as $\text{Gamma}(\alpha, \lambda)$.

The gamma function, $\Gamma(\alpha)$, appears in the denominator of this PDF. The definition and properties of this function are given on page 5 of the *Tables*.

The mean and variance of X are:

$$E(X) = \frac{\alpha}{\lambda}$$

$$\text{var}(X) = \frac{\alpha}{\lambda^2}$$



Question

If $X \sim \text{Gamma}(\alpha, \lambda)$, show that the MGF of X is:

$$M_X(t) = \left(1 - \frac{t}{\lambda}\right)^{-\alpha}$$

Solution

Using the definition of the MGF, we have:

$$M_X(t) = E(e^{tX}) = \int_0^\infty e^{tx} \frac{1}{\Gamma(\alpha)} \lambda^\alpha x^{\alpha-1} e^{-\lambda x} dx = \int_0^\infty \frac{1}{\Gamma(\alpha)} \lambda^\alpha x^{\alpha-1} e^{-(\lambda-t)x} dx$$

We can make the integrand look like the PDF of the $\text{Gamma}(\alpha, \lambda - t)$ distribution by writing:

$$M_X(t) = \left(\frac{\lambda}{\lambda - t}\right)^\alpha \int_0^\infty \frac{1}{\Gamma(\alpha)} (\lambda - t)^\alpha x^{\alpha-1} e^{-(\lambda-t)x} dx$$

This integral is equal to 1 provided $\lambda - t > 0$, so:

$$M_X(t) = \left(\frac{\lambda}{\lambda - t}\right)^\alpha = \left(\frac{\lambda - t}{\lambda}\right)^{-\alpha} = \left(1 - \frac{t}{\lambda}\right)^{-\alpha}, \quad \text{for } t < \lambda$$

By differentiating the MGF, we can obtain the non-central moments, $E(X^k)$, $k = 1, 2, 3, \dots$:

$$E(X) = M'_X(0), \quad E(X^2) = M''_X(0), \quad \text{etc}$$

The variance and skewness can be obtained more quickly using the cumulant generating function (CGF). Recall that:

$$C_X(t) = \ln M_X(t)$$



Question

Suppose that $X \sim \text{Gamma}(\alpha, \lambda)$.

Derive formulae for the skewness and coefficient of skewness of X .

Solution

The skewness of X is its third central moment, $E[(X - E(X))^3]$. It can be obtained by differentiating the CGF three times and evaluating the third derivative when $t = 0$.

Since $X \sim \text{Gamma}(\alpha, \lambda)$:

$$C_X(t) = \ln\left(1 - \frac{t}{\lambda}\right)^{-\alpha} = -\alpha \ln\left(1 - \frac{t}{\lambda}\right)$$

Differentiating using the chain rule:

$$C'_X(t) = -\alpha \left(-\frac{1}{\lambda}\right) \left(1 - \frac{t}{\lambda}\right)^{-1} = \frac{\alpha}{\lambda} \left(1 - \frac{t}{\lambda}\right)^{-1}$$

$$C''_X(t) = \frac{\alpha}{\lambda} \left(-\frac{1}{\lambda}\right) (-1) \left(1 - \frac{t}{\lambda}\right)^{-2} = \frac{\alpha}{\lambda^2} \left(1 - \frac{t}{\lambda}\right)^{-2}$$

$$C'''_X(t) = \frac{\alpha}{\lambda^2} \left(-\frac{1}{\lambda}\right) (-2) \left(1 - \frac{t}{\lambda}\right)^{-3} = \frac{2\alpha}{\lambda^3} \left(1 - \frac{t}{\lambda}\right)^{-3}$$

So:

$$\text{skew}(X) = C'''_X(0) = \frac{2\alpha}{\lambda^3} \left(1 - \frac{0}{\lambda}\right)^{-3} = \frac{2\alpha}{\lambda^3}$$

The coefficient of skewness of X is:

$$\text{coeff of skew}(X) = \frac{\text{skew}(X)}{[\text{var}(X)]^{3/2}}$$

The variance can also be obtained from the CGF:

$$\text{var}(X) = C''_X(0) = \frac{\alpha}{\lambda^2} \left(1 - \frac{0}{\lambda}\right)^{-2} = \frac{\alpha}{\lambda^2}$$

So:

$$\text{coeff of skew}(X) = \frac{\text{skew}(X)}{[\text{var}(X)]^{3/2}} = \frac{2\alpha / \lambda^3}{(\alpha / \lambda^2)^{3/2}} = \frac{2}{\alpha^{1/2}} = \frac{2}{\sqrt{\alpha}}$$

Formulae for the PDF, MGF, mean, variance, non-central moments and coefficient of skewness of the gamma distribution are all given on page 12 of the *Tables*.

There is no closed form (*ie* no simple formula) for the CDF of a gamma random variable, which means that it is not easy to find gamma probabilities directly without using a computer package such as R. However, these probabilities can be obtained using the relationship between the gamma and chi-squared distributions.



Relationship between gamma and chi-squared distributions

If $X \sim \text{Gamma}(\alpha, \lambda)$ and 2α is an integer, then:

$$2\lambda X \sim \chi^2_{2\alpha}$$

This result is also given on page 12 of the *Tables*.

As an illustration of how this relationship can be used, suppose that $X \sim \text{Gamma}(10, 4)$ and we want to calculate $P(X > 4.375)$. Using the result above, we know that $8X \sim \chi^2_{20}$, so:

$$P(X > 4.375) = P(8X > 8 \times 4.375) = P(\chi^2_{20} > 35)$$

From page 166 of the *Tables*, we see that:

$$P(\chi^2_{20} \leq 35) = 0.9799$$

So:

$$P(X > 4.375) = 1 - 0.9799 = 0.0201$$



The R code for simulating a random sample of 100 values from the gamma distribution with $\alpha = 2$ and $\lambda = 0.25$ is:

```
rgamma(100, 2, 0.25)
```

Similarly, the PDF, CDF and quantiles can be obtained using the R functions `dgamma`, `pgamma` and `qgamma`.

1.3 The normal distribution

The normal distribution arises in a variety of contexts. It is of limited use for modelling loss distributions because of its symmetry (as loss distributions tend to be positively skewed).



Question

Derive the formula for the MGF of a standard normal random variable.

Solution

Suppose that $X \sim N(0,1)$. Then the PDF of X is:

$$f(x) = \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}x^2}$$

and the MGF is:

$$M_X(t) = E(e^{tX}) = \int_{-\infty}^{\infty} e^{tx} \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}x^2} dx = \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}(x^2 - 2tx)} dx$$

Completing the square gives:

$$M_X(t) = e^{\frac{1}{2}t^2} \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}(x-t)^2} dx$$

The integrand in the expression immediately above is the PDF of a $N(t,1)$ random variable.

Integrating this over all possible values of x gives the total probability, which is 1. So:

$$M_X(t) = e^{\frac{1}{2}t^2}$$

2 Other loss distributions

The distributions given in Section 1 (exponential, gamma, normal) all have easily derivable MGFs. However, there is a wide variety of other distributions that may also be used to model losses. We consider some of these here. None of the distributions in this section have an MGF that is easy to derive or use.

2.1 The lognormal distribution

The definition of the lognormal distribution is very simple: X has a lognormal distribution if $\log X$ has a normal distribution. When $\log X \sim N(\mu, \sigma^2)$, $X \sim \log N(\mu, \sigma^2)$.

So the range of values taken by the lognormal distribution is 0 to ∞ .

As usual, $\log$ here refers to the natural logarithm, *ie* log base e .

The mean and variance of a lognormal random variable can be obtained from the MGF of the corresponding normal distribution. If $X \sim \log N(\mu, \sigma^2)$, then:

$$E(X) = E(e^{\ln X}) = E(e^Y)$$

where $Y = \ln X \sim N(\mu, \sigma^2)$. However:

$$E(e^Y) = M_Y(1) = e^{\mu + \frac{1}{2}\sigma^2}$$

So $E(X) = e^{\mu + \frac{1}{2}\sigma^2}$.

Similarly:

$$E(X^2) = E(e^{2Y}) = M_Y(2) = e^{2\mu + 2\sigma^2}$$

and hence:

$$\text{var}(X) = e^{2\mu + 2\sigma^2} - \left(e^{\mu + \frac{1}{2}\sigma^2} \right)^2 = e^{2\mu + 2\sigma^2} - e^{2\mu + \sigma^2} = e^{2\mu + \sigma^2} (e^{\sigma^2} - 1)$$

Alternatively, the mean and variance can be derived using integration.

Formulae for the PDF, mean, variance, non-central moments and coefficient of skewness of the lognormal distribution are given on page 14 of the *Tables*.

Lognormal probabilities can be evaluated by expressing them as standard normal probabilities and looking up the values given on pages 160 and 161 of the *Tables*.



The R code for simulating values and obtaining the PDF, CDF and quantiles from the lognormal distribution is similar to the R code used for other continuous distributions using the R functions `rlnorm`, `dlnorm`, `plnorm` and `qlnorm`.

2.2 The two-parameter Pareto distribution

A random variable X has the Pareto distribution with parameters $\alpha > 0$ and $\lambda > 0$ if it has CDF:

$$F(x) = 1 - \left(\frac{\lambda}{\lambda + x} \right)^\alpha, \quad x > 0$$

In that case we write $X \sim Pa(\alpha, \lambda)$.

It is easily checked by differentiating $F(x)$ with respect to x that the Pareto distribution has PDF:

$$f(x) = \frac{\alpha \lambda^\alpha}{(\lambda + x)^{\alpha+1}}, \quad x > 0$$



Question

Suppose that $X \sim Pa(\alpha, \lambda)$. Derive a formula for $E(X)$.

Solution

The expected value is:

$$E(X) = \int_0^\infty x f(x) dx = \int_0^\infty x \frac{\alpha \lambda^\alpha}{(\lambda + x)^{\alpha+1}} dx$$

One way to simplify this is to use the substitution $t = \lambda + x$. Using this substitution:

$$E(X) = \int_\lambda^\infty (t - \lambda) \frac{\alpha \lambda^\alpha}{t^{\alpha+1}} dt = \alpha \lambda^\alpha \int_\lambda^\infty t^{-\alpha} dt - \alpha \lambda^{\alpha+1} \int_\lambda^\infty t^{-\alpha-1} dt$$

Integrating gives:

$$E(X) = \alpha \lambda^\alpha \left[\frac{t^{-\alpha+1}}{-\alpha+1} \right]_\lambda^\infty - \alpha \lambda^{\alpha+1} \left[\frac{t^{-\alpha}}{-\alpha} \right]_\lambda^\infty = \frac{\alpha \lambda}{\alpha-1} - \lambda = \frac{\lambda}{\alpha-1}$$

This expression is valid only if the powers in the bracketed terms are both negative, i.e. if $\alpha > 1$.

Alternatively, this formula could be derived using integration by parts.

Formulae for the CDF, PDF, mean, variance, non-central moments and coefficient of skewness of the Pareto distribution are given on page 14 of the *Tables*.

Let's now consider the median of the Pareto distribution. By definition, the median m is the point where $F(m) = P(X \leq m) = \frac{1}{2}$. So, in this case, we have:

$$1 - \left(\frac{\lambda}{\lambda + m} \right)^\alpha = \frac{1}{2}$$

and this can be rearranged to give:

$$m = \lambda(2^{1/\alpha} - 1)$$

We can compare the median and the mean by drawing a graph. We have just seen that the mean, μ , is equal to $\frac{\lambda}{\alpha - 1}$.

A sketch of the graphs of $\frac{m}{\lambda}$ (bottom curve) and $\frac{\mu}{\lambda}$ (top curve) for values of $\alpha > 1$ is shown below:



From this we see that the mean is always greater than the median, *ie* the Pareto distribution is always positively skewed.



There is no built in R code for the Pareto distribution so we would have to define the functions `rpareto`, `dpareto`, `ppareto` and `qpareto` from first principles as follows:

```
rpareto <- function(n,a,l){
  rp <- l*((1-runif(n))^(1/a)-1)
  rp}

dpareto <- function(x,a,l){
  a*l^(a)/((1+x)^(a+1))}

ppareto <- function(q,a,l){
  1-(1/(1+q))^a}

qpareto <- function(p,a,l){
  q <- l*((1-p)^(1/a)-1)
  q}
```

2.3 The Burr distribution

The CDF of the Pareto distribution $Pa(\alpha, \lambda)$ is:

$$F(x) = 1 - \frac{\lambda^\alpha}{(\lambda + x)^\alpha}, \quad x > 0$$

A further parameter $\gamma > 0$ can be introduced by setting:

$$F(x) = 1 - \frac{\lambda^\alpha}{(\lambda + x^\gamma)^\alpha}, \quad x > 0$$

This is the CDF of the transformed Pareto or Burr distribution. The additional parameter gives extra flexibility when a fit to data is required.

Formulae for the CDF, PDF and non-central moments of the Burr distribution are given on page 15 of the *Tables*.



There is no built in R code for the Burr distribution so we would have to define the functions `rburr`, `dburr`, `pburr` and `qburrr` from first principles as follows:

```
rburr <- function(n,a,l,g) {
  rp <- (1*((1-runif(n))^(1/a)-1))^(1/g)
  rp}

dburr <- function(x,a,l,g) {
  ((a*g*l^a)*x^(g-1))/((1+x^g)^(a+1))}

pburr <- function(q,a,l,g) {
  1-(1/(1+q^g))^a}

qburrr <- function(p,a,l,g) {
  q <- (1*((1-p)^(1/a)-1))^(1/g)
  q}
```

2.4 The three-parameter Pareto distribution

The PDF of the Pareto distribution $Pa(\alpha, \lambda)$ is:

$$f(x) = \frac{\alpha \lambda^\alpha}{(\lambda + x)^{\alpha+1}}, \quad x > 0$$

Another generalisation of the Pareto distribution is to add a further parameter k so that the PDF becomes:

$$f(x) = \frac{\Gamma(\alpha + k) \delta^\alpha}{\Gamma(\alpha) \Gamma(k)} \frac{x^{k-1}}{(\delta + x)^{\alpha+k}}, \quad x > 0$$

Formulae for the PDF, mean, variance and non-central moments of the three-parameter Pareto distribution are given on page 15 of the *Tables*.

The three-parameter Pareto distribution is equivalent to the two-parameter Pareto distribution when $k = 1$.

The moments of the generalised Pareto can be obtained either directly by evaluating

$$E(X^n) = \int_x x^n f(x) dx \text{ or by using a conditional expectation argument.}$$

Here the Core Reading is using the phrase ‘generalised Pareto distribution’ to refer to the three-parameter Pareto distribution. However, this is not the same as the generalised Pareto distribution that we will meet in [Chapter 16](#).

The easiest way to evaluate the integral expression:

$$\int_x x^n f(x) dx$$

is to make it look like the PDF of another three-parameter Pareto distribution.



Question

Suppose that X has a three-parameter Pareto distribution with parameters α , λ and k . Derive formulae for $E(X)$ and $\text{var}(X)$.

Solution

The mean is:

$$E(X) = \int_0^\infty x \frac{\Gamma(\alpha+k)\lambda^\alpha x^{k-1}}{\Gamma(\alpha)\Gamma(k)(\lambda+x)^{\alpha+k}} dx = \int_0^\infty \frac{\Gamma(\alpha+k)\lambda^\alpha x^k}{\Gamma(\alpha)\Gamma(k)(\lambda+x)^{\alpha+k}} dx$$

This expression can be simplified by making the integrand look like the PDF of the Pareto distribution with parameters $\alpha-1$, λ and $k+1$:

$$E(X) = \lambda \frac{\Gamma(\alpha-1)}{\Gamma(\alpha)} \frac{\Gamma(k+1)}{\Gamma(k)} \int_0^\infty \frac{\Gamma(\alpha+k)\lambda^{\alpha-1} x^k}{\Gamma(\alpha-1)\Gamma(k+1)(\lambda+x)^{\alpha+k}} dx$$

Since this integrand is a PDF, integrating it over all possible values of x gives us 1. So:

$$E(X) = \lambda \frac{\Gamma(\alpha-1)}{\Gamma(\alpha)} \frac{\Gamma(k+1)}{\Gamma(k)} = \frac{\lambda k}{\alpha-1}$$

Here we are using the result $\Gamma(\alpha) = (\alpha-1)\Gamma(\alpha-1)$, which is given on page 5 of the *Tables*.

We can use a similar method to calculate the second moment:

$$E(X^2) = \int_0^\infty x^2 \frac{\Gamma(\alpha+k)\lambda^\alpha x^{k-1}}{\Gamma(\alpha)\Gamma(k)(\lambda+x)^{\alpha+k}} dx = \int_0^\infty \frac{\Gamma(\alpha+k)\lambda^\alpha x^{k+1}}{\Gamma(\alpha)\Gamma(k)(\lambda+x)^{\alpha+k}} dx$$

Making the integrand look like the PDF of the Pareto distribution with parameters $\alpha - 2$, λ and $k + 2$, we have:

$$E(X^2) = \lambda^2 \frac{\Gamma(\alpha - 2)}{\Gamma(\alpha)} \frac{\Gamma(k + 2)}{\Gamma(k)} \int_0^\infty \frac{\Gamma(\alpha + k) \lambda^{\alpha - 2} x^{k + 1}}{\Gamma(\alpha - 2) \Gamma(k + 2) (\lambda + x)^{\alpha + k}} dx = \frac{k(k + 1) \lambda^2}{(\alpha - 1)(\alpha - 2)}$$

So the variance is:

$$\begin{aligned} \frac{\lambda^2 k(k + 1)}{(\alpha - 1)(\alpha - 2)} - \left(\frac{\lambda k}{\alpha - 1} \right)^2 &= \frac{\lambda^2 k(k + 1)(\alpha - 1) - \lambda^2 k^2 (\alpha - 2)}{(\alpha - 1)^2 (\alpha - 2)} \\ &= \frac{\lambda^2 k [(k\alpha + \alpha - k - 1) - (k\alpha - 2k)]}{(\alpha - 1)^2 (\alpha - 2)} \\ &= \frac{\lambda^2 k(k + \alpha - 1)}{(\alpha - 1)^2 (\alpha - 2)} \end{aligned}$$



There is no built in R code for the three-parameter Pareto distribution, so we would have to define the function `dgporeto` from first principles as we did for the Pareto. However, since the CDF does not exist in closed form it is not easy to create functions to obtain probabilities, percentage points or simulated values.

2.5 The Weibull distribution

The Pareto distribution is a distribution with an upper tail that tends to 0 as a power of x . This gives a distribution with a much heavier tail than the exponential. The expressions for the upper tails of the exponential and the Pareto distributions are:

exponential $P(X > x) = \exp(-\lambda x)$

Pareto $P(X > x) = (\lambda / (\lambda + x))^\alpha$

So, if we want to choose a model with a thick tail so as not to underestimate the probability of a large claim, we might well choose the Pareto distribution to model our claims (assuming that it is a suitable distribution in other respects).

However, these are not the only types of tail.

There is a further possibility. Set:

$$P(X > x) = \exp(-\lambda x^\gamma), \quad \gamma > 0$$

There are now two cases. If $\gamma < 1$, a distribution with a tail intermediate in weight between the exponential and the Pareto will be obtained, while if $\gamma > 1$, the upper tail will be lighter than the exponential ($\gamma = 1$ is the exponential distribution).

This distribution is called the Weibull distribution, a very flexible distribution, which can be used as a model for losses in insurance, usually with $\gamma < 1$. A random variable X has a Weibull distribution with parameters $c > 0$ and $\gamma > 0$ if it has CDF:

$$F(x) = 1 - \exp(-cx^\gamma), \quad x > 0$$

In that case we write $X \sim W(c, \gamma)$. (Note the change from λ to c ; this is the notation used in the *Tables for Actuarial Examinations*).

The PDF of the $W(c, \gamma)$ distribution is:

$$f(x) = c\gamma x^{\gamma-1} \exp(-cx^\gamma), \quad x > 0$$

Formulae for the CDF, PDF and non-central moments of the Weibull distribution are given on page 15 of the *Tables*.



Question

Suppose that X has a Weibull distribution with parameters c and γ . Derive a formula for $E(X)$.

Solution

The mean is:

$$E(X) = \int_0^{\infty} x c\gamma x^{\gamma-1} e^{-cx^\gamma} dx$$

Making the substitution $u = cx^\gamma$, so that $\frac{du}{dx} = c\gamma x^{\gamma-1}$ and $x = \left(\frac{u}{c}\right)^{1/\gamma}$ gives:

$$E(X) = \int_0^{\infty} \left(\frac{u}{c}\right)^{1/\gamma} e^{-u} du$$

Now, manipulating the integrand so that it looks like the PDF of a $\text{Gamma}\left(1 + \frac{1}{\gamma}, 1\right)$ random variable, we have:

$$E(X) = \Gamma\left(1 + \frac{1}{\gamma}\right) \frac{1}{c^{1/\gamma}} \int_0^{\infty} \frac{1}{\Gamma(1 + 1/\gamma)} u^{1/\gamma} e^{-u} du$$

The integral is now equal to 1 (as we're integrating a PDF over all possible values of the random variable). So:

$$E(X) = \Gamma\left(1 + \frac{1}{\gamma}\right) \frac{1}{c^{1/\gamma}}$$



The R code for simulating a random sample of 100 values from the Weibull distribution with $c = 2$ and $\gamma = 0.25$ is:

```
rweibull(100, 0.25, 2^(-1/0.25))
```

R uses a different parameterisation for the scale parameter, c .

Similarly, the PDF, CDF and quantiles can be obtained using the R functions `dweibull`, `pweibull` and `qweibull`.

Alternatively, we could redefine them from first principles as follows:

```
rweibull <- function(n,c,g) {
  rp <- (log(1-runif(n))/c)^(1/g)
  rp}

dweibull <- function(x,c,g) {
  c*g*x^(g-1)*exp(-c*x^g)}

pweibull <- function(q,c,g) {
  1-exp(-c*x^g)}

qweibull <- function(p,c,g) {
  q <- (log(1-p)/c)^(1/g)
  q}
```

2.6 Illustration of tail weights

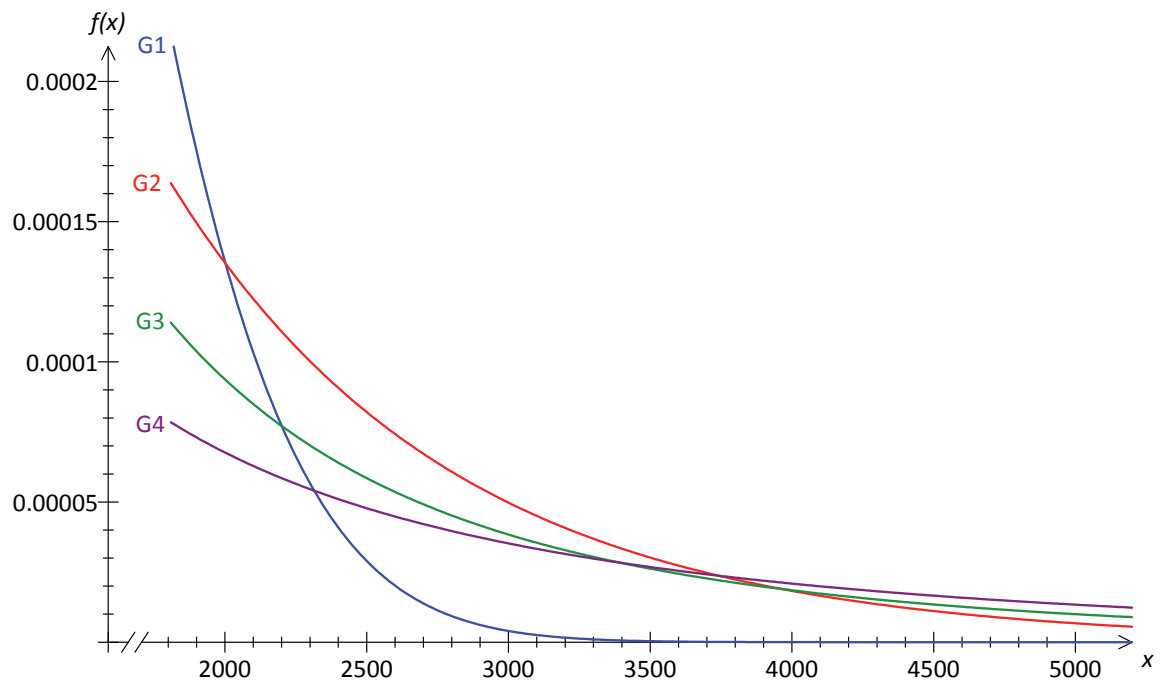
The PDFs shown in the diagram below illustrate the difference in the tails of the exponential, Pareto and Weibull distributions. All four of the distributions have a mean of 1,000.

G1 is a Weibull distribution with parameters $\gamma = 2$ and $c = \frac{\pi}{2,000^2} = 7.854 \times 10^{-7}$ (standard deviation = 522.7).

G2 is an exponential distribution with $\lambda = \frac{1}{1,000} = 0.001$ (standard deviation = 1,000).

G3 is a (two-parameter) Pareto distribution with $\alpha = 3$ and $\lambda = 2,000$ (standard deviation = 1,732).

G4 is a Weibull distribution with $\gamma = \frac{1}{2}$ and $c = \frac{1}{\sqrt{500}} = 0.04472$ (standard deviation = 2,236).



3 Estimation

The methods of maximum likelihood, moments and percentiles can be used to fit distributions to sets of data.



We can check the fit in R by plotting a histogram of the data and superimposing the density function of the fitted distribution. Better yet, we can plot an empirical density function from the data using the function `density` and add the true density function of the fitted distribution.

A better way is to use the `qqplot` function to compare the sample data to simulated values from the fitted model distribution. A straight diagonal line indicates perfect fit:

```
qqplot(<simulated theoretical values>, <sample values>)
abline(0,1)
```

The fit of the distribution can also be tested formally by using a χ^2 test. The method of percentiles is outlined in Section 3.3; the other methods and the χ^2 test have been covered in Subject CS1, Actuarial Statistics 1.

We will now give a summary of the method of moments and maximum likelihood estimation. We will also introduce the method of percentiles and give a brief reminder of how the chi-squared test can be used to check the fit of a statistical distribution to a data set.

3.1 The method of moments

For a distribution with r parameters, the moments are as follows:

$$m_j = \frac{1}{n} \sum_{i=1}^n x_i^j \quad j = 1, 2 \dots r$$

where:

$m_j = E(X^j | \theta)$, a function of the unknown parameter, θ , being estimated

n = the sample size

x_i = the i th value in the sample

The estimate for the parameter, θ , can be determined by solving the equation above. Where there is more than one parameter, they can be determined by solving the simultaneous equations for each m_j .

So, for example, if we are trying to estimate the value of a single parameter, and we have a sample of n claims whose sizes are $x_1, x_2, \dots, x_n$, we would solve the equation:

$$E(X) = \frac{1}{n} \sum_{i=1}^n x_i$$

ie we would equate the first non-central moments for the population and the sample.

If we are trying to find estimates for two parameters (for example if we are fitting a gamma distribution and need to obtain estimates for both parameters), we would solve the simultaneous equations:

$$E(X) = \frac{1}{n} \sum_{i=1}^n x_i \quad \text{and} \quad E(X^2) = \frac{1}{n} \sum_{i=1}^n x_i^2$$

In fact, in the two-parameter case, estimates are often obtained by equating sample and population means and variances. If we use the n -denominator sample variance:

$$s_n^2 = \frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^2 = \frac{1}{n} \left[\sum_{i=1}^n x_i^2 - n\bar{x}^2 \right]$$

this will give the same estimates as would be obtained by equating the first two non-central moments.

More generally, we use as many equations of the form $E(X^k) = \frac{1}{n} \sum_{i=1}^n x_i^k$, $k = 1, 2, \dots$ as are needed to determine estimates of the relevant parameters.

3.2 Maximum likelihood estimation

The likelihood function of a random variable, X , is the probability (or PDF) of observing what was observed given a hypothetical value of the parameter, θ . The maximum likelihood estimate (MLE) is the one that yields the highest probability (or PDF), ie that maximises the likelihood function.

For the sample in Section 3.1 above, the likelihood function $L(\theta)$ can be expressed as:

$$L(\theta) = \prod_{i=1}^n P(X = x_i | \theta) \quad \text{for a discrete random variable, } X$$

or:

$$L(\theta) = \prod_{i=1}^n f(x_i | \theta) \quad \text{for a continuous random variable, } X$$

To determine the MLE the likelihood function needs to be maximised. Often it is practical to consider the log-likelihood function:

$$l(\theta) = \log L(\theta) = \sum_{i=1}^n \log P(X = x_i | \theta) \quad \text{for a discrete random variable, } X$$

or:

$$l(\theta) = \log L(\theta) = \sum_{i=1}^n \log f(x_i | \theta) \quad \text{for a continuous random variable, } X$$

If $l(\theta)$ can be differentiated with respect to θ , the MLE, expressed as $\hat{\theta}$, satisfies the expression:

$$\frac{d}{d\theta} l(\hat{\theta}) = 0$$

Where there is more than one parameter, the MLEs for each parameter can be determined by taking partial derivatives of the log-likelihood function and setting each to zero.

The determination of MLEs when the data are incomplete is covered in [Chapter 18](#).

We will now look at the distributions described earlier in this chapter and consider how the parameters can be estimated in each case.

The exponential distribution

It is possible to use the method of maximum likelihood (ML) or the method of moments to estimate the parameter of the exponential distribution.

For example, suppose that an insurance company uses an exponential distribution to model the cost of repairing insured vehicles that are involved in accidents, and the average cost of repairing a random sample of 1,000 vehicles is £2,200.

We can calculate the maximum likelihood estimate of the exponential parameter as follows.

Let $x_1, x_2, \dots, x_{1,000}$ denote the individual repair costs.

The likelihood of obtaining these values for the costs, if they come from an exponential distribution with parameter λ , is:

$$L = \prod_{i=1}^{1,000} \lambda e^{-\lambda x_i} = \lambda^{1,000} e^{-\lambda \sum x_i} = \lambda^{1,000} e^{-1,000 \lambda \bar{x}}$$

(where $\bar{x} = \frac{1}{1,000} \sum_{i=1}^{1,000} x_i$ denotes the average claim amount).

We want to determine the value of λ that maximises the likelihood, or equivalently the value that maximises the log-likelihood:

$$\ln L = 1,000 \ln \lambda - 1,000 \lambda \bar{x}$$

Differentiating with respect to λ :

$$\frac{\partial}{\partial \lambda} \ln L = \frac{1,000}{\lambda} - 1,000 \bar{x}$$

This is equal to 0 when:

$$\lambda = \frac{1}{\bar{x}}$$

The second derivative is:

$$\frac{\partial^2}{\partial \lambda^2} \log L = -\frac{1,000}{\lambda^2}$$

Since the second derivative is negative when $\lambda = \frac{1}{\bar{x}}$, the stationary point is a maximum. So, $\hat{\lambda}$, the maximum likelihood estimate of λ is $\frac{1}{\bar{x}}$, or $\frac{1}{2,200}$.

Alternatively, we could argue that the likelihood function is continuous and is always positive (by necessity) and that $\lambda^n e^{-\lambda n \bar{x}} \rightarrow 0$ as $\lambda \rightarrow 0$ or $\lambda \rightarrow \infty$. So any stationary point that we find must be a maximum.



To obtain ML estimates in R, we could use the `fitdistr` in the MASS package as follows:

```
fitdistr(<data vector>, "exponential")
```

Or we could define the log-likelihood function and use the function `nlm` on the negative value of the log-likelihood function.

```
nlm(-<log likelihood function>, <vector of parameters>)
```

So to fit an exponential distribution to a vector `<data>` with initial estimate of $\lambda = 0.5$ we would use:

```
params <- 0.5
n <- length(<data>)
sx <- sum(<data>)
fMLE <- function(params) {n*log(params[1]) - params[1]*sx}
nlm(-fMLE, params)
```

The gamma distribution

The moments have a simple form and so the method of moments is very easy to apply. The MLEs for the gamma distribution cannot be obtained in closed form (*ie* in terms of elementary functions) but the moment estimators can be used as initial estimators in the search for the MLEs.

It is more convenient to obtain MLEs for the gamma distribution using a different parameterisation. Set $\mu = \alpha / \lambda$ and estimate the parameters α and μ . Then recover the MLE of λ by setting $\hat{\lambda} = \hat{\alpha} / \hat{\mu}$. This uses the invariance property of maximum likelihood estimators.

The invariance property says that if $\hat{\theta}$ is the maximum likelihood estimator of θ and $f(\theta)$ is a function of θ , then $f(\hat{\theta})$ is the maximum likelihood estimator of $f(\theta)$.

For the gamma distribution, λ is a function of both α and μ .



To obtain ML estimates in R, we could use the `fitdistr` in the MASS package as follows:

```
fitdistr(<data vector>, "gamma")
```

However, it is better to include the initial estimates obtained from the method of moments (and put a lower limit of say, $0.001 > 0$, to prevent invalid answers). For example:

```
fitdistr(<data vector>, dgamma, list(shape = <alpha>,
rate = <lambda>), lower = 0.001)
```

Alternatively, we could define the log-likelihood function and use the function `nlm` on the negative value of the log-likelihood function as before.

The normal distribution

The method of moments and maximum likelihood estimation are both straightforward to apply in this case. Both give the following estimates:

$$\hat{\mu} = \bar{x} \quad \text{and} \quad \hat{\sigma}^2 = \frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^2$$

The estimate for the population variance is $\frac{n-1}{n} \times$ the usual sample variance. Of course, provided the sample size is large, there will be little difference between estimates calculated using the two different sample variance formulae.

The lognormal distribution

Estimation for the lognormal distribution is straightforward since μ and σ^2 may be estimated using the log-transformed data. Let $x_1, x_2, \dots, x_n$ be the observed values and let $y_i = \log x_i$. The MLEs of μ and σ^2 are $\bar{y}$ and s_y^2 , where the subscript y signifies a sample variance (n -denominator) computed on the y values.

In other words, the maximum likelihood estimate of σ^2 is:

$$\hat{\sigma}^2 = \frac{1}{n} \sum_{i=1}^n (y_i - \bar{y})^2$$

Alternatively the method of moments can be used to estimate the parameters.

As an example, suppose that based on an analysis of past claims, an insurance company believes that individual claims in a particular category for the coming year will be lognormally distributed with a mean size of £5,000 and a standard deviation of £7,500. The company wants to estimate the proportion of claims that will exceed £25,000.

To do this, it needs to estimate the parameters, μ and σ^2 , of the lognormal distribution. Equating the formulae for the mean and standard deviation of the lognormal distribution to the values given gives:

$$e^{\mu + \frac{1}{2}\sigma^2} = 5,000 \quad \text{and} \quad e^{\mu + \frac{1}{2}\sigma^2} \sqrt{e^{\sigma^2} - 1} = 7,500$$

Dividing the second equation by the first gives:

$$\sqrt{e^{\sigma^2} - 1} = \frac{7,500}{5,000} = 1.5$$

$$\Rightarrow \sigma^2 = 1.179$$

We can now solve for μ :

$$\mu = \log 5,000 - \frac{1}{2}(1.179) = 7.928$$

So the proportion of claims expected to exceed £25,000 is:

$$\begin{aligned} P(X > 25,000) &= P(\ln X > \ln 25,000) \\ &= P(N(7.928, 1.179) > \ln 25,000) \\ &= P\left(N(0, 1) > \frac{\ln 25,000 - 7.928}{\sqrt{1.179}}\right) \\ &= 1 - \Phi(2.025) = 0.021 \end{aligned}$$

ie 2.1% of claims are expected to exceed £25,000.



To obtain ML estimates in R, we could use the `fitdistr` in the MASS package as follows:

```
fitdistr(<data vector>, "log-normal")
```

Alternatively, we could define the log-likelihood function and use the function `nlm` on the negative value of the log-likelihood function as before.

The two-parameter Pareto distribution

The method of moments is very easy to apply in the case of the two-parameter Pareto distribution, but the estimators obtained in this way will tend to have rather large standard errors, mainly because S^2 , the sample variance, has a very large variance. However, the method does provide initial estimates for more efficient methods of estimation that may not be so simple to apply, like maximum likelihood, where numerical methods may need to be used.



Question

Claims arising from a particular group of policies are believed to follow a Pareto distribution with parameters α and λ . A random sample of 20 claims gives values such that $\sum x = 1,508$ and $\sum x^2 = 257,212$. Estimate α and λ using the method of moments.

Solution

Suppose that X is the claim amount random variable. Then:

$$E(X) = \frac{\lambda}{\alpha - 1}$$

Rearranging the variance formula to find $E(X^2)$, we have:

$$E(X^2) = \text{var}(X) + [E(X)]^2 = \frac{2\lambda^2}{(\alpha - 1)(\alpha - 2)}$$

So we set:

$$\frac{\lambda}{\alpha - 1} = \frac{1,508}{20} = 75.4 \quad \text{and} \quad \frac{2\lambda^2}{(\alpha - 1)(\alpha - 2)} = \frac{257,212}{20} = 12,860.6$$

Squaring the first of these equations and substituting into the second, we see that:

$$\frac{2 \times 75.4^2 (\alpha - 1)}{\alpha - 2} = 12,860.6$$

Solving this equation, we find that the method of moments estimates of α and λ are 9.630 and 650.7, respectively.

The three-parameter Pareto distribution

Things are not quite so easy for the three-parameter Pareto distribution.

As for estimation, the CDF does not exist in closed form, so the method of percentiles is not available.

The method of percentiles is described in Section 3.3.

ML can be used, but again suitable computer software is required; the method of moments can provide initial estimates for any iterative scheme.



We will need to define the log-likelihood function and use the function `nlm` on the negative value of the log-likelihood function as before.

The Weibull and Burr distributions

Neither the method of moments nor maximum likelihood is elementary to apply if both c and γ are unknown (although if a computer is available, as would be the case in practice, the equations are simple enough).



To obtain ML estimates in R, we could use the `fitdistr` in the MASS package as follows:

```
fitdistr(<data vector>,"weibull")
```

However, it is better to include the initial estimates obtained from the method of percentiles (and put a lower limit of say, $0.001 > 0$, to prevent invalid answers). For example:

```
fitdistr(<data vector>,dweibull,
list(shape = <gamma>, scale = <c^(-1/gamma)>, lower = 0.001)
```

Alternatively, we could define the log-likelihood function and use the function `nlm` on the negative value of the log-likelihood function as before.

In the case where γ has the known value γ^* , maximum likelihood is easy enough.

To do this, we let $y_i = x_i^{\gamma}$. If the original distribution is Weibull, the y values now have an exponential distribution. If the original distribution is Burr, the y values now come from a Pareto distribution. This can be seen by comparing the CDFs.

In the case of the Weibull, the MLE of c can now be determined in the usual way. In the case of the Burr distribution, the estimates have to be calculated numerically (since the MLEs of the parameters of the Pareto distribution cannot be calculated algebraically).

3.3 The method of percentiles

The distribution function of the $W(c, \gamma)$ distribution is an elementary function, and a simple method of estimation of both c and γ is based on this. The method involves equating selected sample percentiles to the distribution function; for example, equate the sample quartiles, the 25th and 75th sample percentiles, to the population quartiles. This corresponds to the way in which sample moments are equated to population moments in the method of moments. This method will be referred to as the method of percentiles.

In the method of moments, the first two moments are used if there are two unknown parameters, and this seems intuitively reasonable (although the theoretical basis for this is not so clear). In a similar fashion, when using the method of percentiles, the median would be used if there were one parameter to estimate. With two parameters, the best procedure is less clear, but the lower and upper quartiles seem a sensible choice.

Example

Estimate c and γ in the Weibull distribution using the method of percentiles, where the first sample quartile is 401 and the third sample quartile is 2,836.75.

Solution

The two equations for c and γ are:

$$F(401) = 1 - \exp(-c \times 401^\gamma) = 0.25$$

$$F(2,836.75) = 1 - \exp(-c \times 2,836.75^\gamma) = 0.75$$

which can be rewritten as:

$$-c \times 401^\gamma = \ln 0.25$$

and: $-c \times 2,836.75^\gamma = \ln 0.75$

Dividing, it is found that $\tilde{\gamma} = 0.8038$, and hence $\tilde{c} = 0.002326$, where $\sim$ denotes the percentile estimate. Note that $\tilde{\gamma}$ is less than 1, indicating a fatter tail than the exponential distribution gives.

We can apply the method of percentiles to any distribution for which it is possible to calculate a closed form for the cumulative distribution function, although the resulting algebra can be messy.



Question

Claims arising from a particular group of policies are believed to follow a Pareto distribution with parameters α and λ . A random sample of 20 claims has a lower quartile of 11 and an upper quartile of 85. Estimate the values of α and λ using the method of percentiles.

Solution

The cumulative distribution function of the Pareto distribution is $F(x) = 1 - \left(\frac{\lambda}{\lambda + x} \right)^\alpha$.

Q_1 , the lower quartile of the distribution, satisfies the equation:

$$F(Q_1) = 1 - \left(\frac{\lambda}{\lambda + Q_1} \right)^\alpha = 0.25$$

and Q_3 , the upper quartile of the distribution, satisfies the equation:

$$F(Q_3) = 1 - \left(\frac{\lambda}{\lambda + Q_3} \right)^\alpha = 0.75$$

So:

$$Q_1 = \lambda \left[(3/4)^{-1/\alpha} - 1 \right] \quad \text{and} \quad Q_3 = \lambda \left[(1/4)^{-1/\alpha} - 1 \right]$$

The method of percentiles estimates $\tilde{\alpha}$ and $\tilde{\lambda}$ are obtained by setting $Q_1 = 11$ and $Q_3 = 85$:

$$11 = \tilde{\lambda} \left[(3/4)^{-1/\tilde{\alpha}} - 1 \right]$$

$$85 = \tilde{\lambda} \left[(1/4)^{-1/\tilde{\alpha}} - 1 \right]$$

We can eliminate $\tilde{\lambda}$ by dividing these equations. This gives:

$$\frac{11}{85} = \frac{(3/4)^{-1/\tilde{\alpha}} - 1}{(1/4)^{-1/\tilde{\alpha}} - 1}$$

We cannot solve this algebraically but it can easily be done on a computer, *eg* using the goalseek function in Excel. Doing this, we find that $\tilde{\alpha} = 1.284$ and hence $\tilde{\lambda} = 43.790$.

These estimates are very different from those obtained using the method of moments. (In an earlier question, we calculated the method of moments estimates of α and λ to be 9.630 and 650.7, respectively.)

The method of percentiles is very unreliable for estimating the parameters of a Pareto distribution unless we use extremely large samples. In this particular case, the method of percentiles is unlikely to give us reasonable estimates unless we use samples of, say, 1,000 or more.

We now turn to the Burr distribution.

Since the CDF exists in closed form, it may be possible to fit the Burr distribution to data by using the method of percentiles; ML will certainly require the use of computer software that allows non-linear optimisation.



We will need to define the log-likelihood function and use the function `nlm` on the negative value of the log-likelihood function as before.

4 Goodness-of-fit tests

As mentioned earlier, one way of testing whether a given loss distribution provides a good model for the observed claim amounts is to apply a chi-squared goodness-of-fit test.

Recall that the formula for the test statistic is $\sum \frac{(O-E)^2}{E}$, where:

- O is the observed number in a particular category
- E is the corresponding expected number predicted by the assumed probabilities
- the sum is over all possible categories.

A high value for the total indicates that the overall discrepancy is quite large and would lead us to reject the model.

As an example, suppose that an insurance company uses an exponential distribution to model the cost of repairing insured vehicles that are involved in accidents, and the average cost of repairing a random sample of 1,000 vehicles is £2,200. A breakdown of the repair costs revealed the following numbers in different bands:

Repair cost, £	Observed number
0 – 1,000	200
1,000 – 2,000	300
2,000 – 3,000	250
3,000 – 4,000	150
4,000 – 5,000	100
5,000+	0

We can use this information to test whether the exponential distribution provides a good model for the individual repair costs.

Here we are testing:

H_0 : repair costs are exponentially distributed

against:

H_1 : repair costs are not exponentially distributed

In order to apply the chi-squared test, we need to calculate the expected number of repair costs in each interval based on the assumption that the null hypothesis is true.

Using the maximum likelihood estimate of the value of λ (ie $1/2,200$), the probability that an individual repair cost will fall in the interval £2,000 - £3,000 is:

$$\int_{2,000}^{3,000} \lambda e^{-\lambda x} dx = \left[-e^{-\lambda x} \right]_{2,000}^{3,000} = e^{-2,000\lambda} - e^{-3,000\lambda} = 0.1472$$

and the expected number for this band is: $1,000 \times 0.1472 = 147.2$

The expected numbers for all the intervals can be calculated in a similar way, giving the following results:

365.3, 231.8, 147.2, 93.4, 59.3, 103.0

The value of the test statistic is:

$$\sum \frac{(O-E)^2}{E} = \frac{(200-365.3)^2}{365.3} + \frac{(300-231.8)^2}{231.8} + \dots + \frac{(0-103.0)^2}{103.0} = 331.89$$

We have 6 intervals, but we have equated the totals and estimated one parameter. So there are $6 - 1 - 1 = 4$ degrees of freedom.

The observed value of the chi-squared statistic far exceeds 14.86, the upper 99.95% point of the chi-squared distribution with 4 degrees of freedom (given on p169 of the *Tables*). So we can reject H_0 with almost total confidence and conclude that the repair costs do not conform to an exponential distribution.

In fact we need only work out the value of the first term in the chi-squared statistic to see that we will reject the null hypothesis.

This conclusion is supported by the observation that, if the values did come from an exponential distribution, we would expect the numbers in each band to decline steadily. However, we recorded 100 fewer values in the first band than in the second.

Chapter 15 Summary

Loss distributions

Individual claim amounts can be modelled using a loss distribution. Loss distributions are often positively skewed and long-tailed.

The (cumulative) distribution function of X is denoted by $F_X(x)$. It is defined by the equation: $F_X(x) = P(X \leq x)$.

The (probability) density function of X is denoted by $f_X(x)$. It is defined by the equation: $f_X(x) = F'_X(x)$, wherever this derivative exists.

Distributions such as the exponential, normal, lognormal, gamma, Pareto, Burr and Weibull distributions are commonly used to model individual claim amounts.

Once the form of the loss distribution has been decided upon, the values of the parameters must be estimated. This might be done using the method of maximum likelihood, the method of moments, or the method of percentiles. Goodness of fit can then be checked using a chi-squared test.

Method of moments

The method of moments involves equating population and sample moments to solve for the unknown parameter values. If there is one parameter to estimate, we equate the population mean with the sample mean. If there are two parameters to estimate, we could equate the first two non-central population moments with the equivalent non-central sample moments. Equivalently, we could equate the first two central population moments with the equivalent central sample moments, noting that (for equivalence) we would need to use the n -denominator sample variance.

Method of maximum likelihood

The steps involved in finding a maximum likelihood estimate (MLE) are as follows:

- write down the likelihood function L – this is the probability/PDF of obtaining the values we have observed
- take logs and simplify the resulting expression
- differentiate the log-likelihood with respect to each parameter to be estimated – this will involve partial differentiation if there is more than one parameter to be estimated
- set the derivatives equal to 0 and solve the equations simultaneously
- check that the resulting values are maxima.

Method of percentiles

The method of percentiles involves equating population and sample percentiles to solve for the unknown parameter values. If there is just one parameter to estimate, we equate the population median with the sample median. If there are two parameters to estimate, we equate the population lower and upper quartiles with the sample lower and upper quartiles.

Testing goodness of fit

We can test whether a given loss distribution provides a good model for the observed claim amounts by applying a chi-squared goodness-of-fit test.

The formula for the test statistic is $\sum \frac{(O - E)^2}{E}$, where:

- O is the observed number in a particular category
- E is the corresponding expected number predicted by the assumed probabilities
- the sum is over all possible categories.

Under the null hypothesis (that the model is correct), the test statistic has a chi-squared distribution.



Chapter 15 Practice Questions

- 15.1 Losses arising from a portfolio follow a Pareto distribution with parameters $\alpha = 3$ and $\lambda = 2,000$.

Calculate the probability that a randomly chosen loss amount exceeds the mean loss amount.

- 15.2 Suppose that X has a Weibull distribution with parameters c and γ .

- (i) Using the formula for $E(X^r)$ given in the *Tables*, write down an expression for $\text{var}(X)$.
- (ii) Show that, when $\gamma = 1$, this reduces to the formula for the variance of an exponential random variable.

- 15.3 Show that if $\gamma = \frac{1}{2}$, the standard deviation of the Weibull distribution is greater than the mean, whereas if $\gamma = 2$ the opposite is true.

- 15.4 The random variable X follows a gamma distribution with parameters $\alpha = 20$ and $\lambda = 0.1$. Determine the value of a such that:

$$P(X > a) = 0.05$$

- 15.5 The random variable X has a Burr distribution with parameters $\gamma = 2$ and $\lambda = 500$.

- (i) Show that the maximum likelihood estimate of the parameter α , based on a random sample $x_1, x_2, \dots, x_n$ is:

$$\hat{\alpha} = \frac{n}{\sum \log(500 + x_i^2) - n \log 500}$$

You may assume that this is a maximum.

- (ii) Evaluate this based on a sample consisting of the five values 52, 109, 114, 163 and 181.

- 15.6 Claim amounts from a particular group of policies have the following distribution:

Amount	£200	£500
Probability	p	$1 - p$

In a random sample of 40 claims, 25 were for £200 and the other 15 were for £500.

Calculate the maximum likelihood estimate of p .

- 15.7 A loss amount random variable has MGF:

$$M(t) = 0.4(1 - 20t)^{-2} + 0.6(1 - 30t)^{-3}$$

Calculate the expected loss amount.

- 15.8** Individual claim amounts on a portfolio of motor insurance policies follow a gamma distribution with parameters α and λ . It is known that $\lambda = 0.8$ for all drivers, but the value of α varies across the population.

Exam style

Given that $\alpha \sim \text{Gamma}(200, 0.5)$, calculate the mean and variance of a randomly chosen claim amount. [5]

- 15.9** (i) The distribution of claims on a portfolio of general insurance policies is a Weibull distribution, with density function $f_1(x)$ where:

Exam style

$$f_1(x) = 2cx e^{-cx^2} \quad (x > 0)$$

It is expected that one claim out of every 100 will exceed £1,000. Use this information to estimate c . [2]

- (ii) An alternative suggestion is that the density function is $f_2(x)$, where:

$$f_2(x) = \lambda e^{-\lambda x} \quad (x > 0)$$

Use the same information as in part (i) to estimate λ . [2]

- (iii) (a) For each of $f_1(x)$ and $f_2(x)$ calculate the value of M such that:

$$P(X > M) = 0.001$$

- (b) Comment on these results. [3]

[Total 7]

- 15.10** A random sample of 100 claim amounts $x_1, x_2, \dots, x_{100}$ is observed from a Weibull distribution with parameter $\gamma = 2$, where c is unknown. For these data:

Exam style

$$\sum x_i = 487,926 \quad \sum x_i^2 = 976,444,000 \quad \text{sample median} = 4,500$$

- (i) Show that the maximum likelihood estimate for c based on a sample of size n is given by:

$$\hat{c} = n / \sum x_i^2$$

and hence estimate the value of c . [4]

- (ii) Estimate the value of c using the method of moments. [2]

- (iii) Calculate the method of percentiles estimate of c . [2]

[Total 8]

- 15.11** Claims arising from a certain type of insurance policy are believed to follow an exponential distribution. The lower quartile claim is 200.

Exam style

Calculate the mean claim size. [3]



Chapter 15 Solutions

- 15.1 Let X denote the loss amount random variable. Then $X \sim Pa(3, 2000)$ and $E(X) = \frac{2,000}{3-1} = 1,000$.

So the required probability is:

$$P(X > 1,000) = 1 - F_X(1,000) = 1 - \left[1 - \left(\frac{2,000}{2,000 + 1,000} \right)^3 \right] = \left(\frac{2,000}{3,000} \right)^3 = 0.29630$$

- 15.2 (i) **Variance**

We have:

$$E(X) = \Gamma\left(1 + \frac{1}{\gamma}\right) \frac{1}{c^{1/\gamma}} \quad \text{and} \quad E(X^2) = \Gamma\left(1 + \frac{2}{\gamma}\right) \frac{1}{c^{2/\gamma}}$$

So:

$$\text{var}(X) = E(X^2) - [E(X)]^2 = \Gamma\left(1 + \frac{2}{\gamma}\right) \frac{1}{c^{2/\gamma}} - \left[\Gamma\left(1 + \frac{1}{\gamma}\right) \frac{1}{c^{1/\gamma}} \right]^2$$

- (ii) **Simplification when $\gamma = 1$**

When $\gamma = 1$, this becomes:

$$\Gamma(3) \times \frac{1}{c^2} - \left[\Gamma(2) \times \frac{1}{c} \right]^2 = \frac{2}{c^2} - \left[\frac{1}{c} \right]^2 = \frac{1}{c^2}$$

which is the formula for the variance of an $\text{Exp}(c)$ random variable.

- 15.3 When $\gamma = \frac{1}{2}$:

$$E(X) = \frac{\Gamma(1+2)}{c^2} = \frac{2!}{c^2} = \frac{2}{c^2}$$

$$E(X^2) = \frac{\Gamma(1+4)}{c^4} = \frac{4!}{c^4} = \frac{24}{c^4}$$

and:

$$\text{var}(X) = E(X^2) - [E(X)]^2 = \frac{24}{c^4} - \left(\frac{2}{c^2} \right)^2 = \frac{20}{c^4}$$

So the standard deviation is $\frac{\sqrt{20}}{c^2}$, which is greater than $E(X)$.

When $\gamma = 2$:

$$E(X) = \frac{\Gamma(1 + \frac{1}{2})}{c^{\frac{1}{2}}} = \frac{\Gamma(1.5)}{c^{\frac{1}{2}}}$$

Using the properties of the gamma function given on page 5 of the *Tables*:

$$\Gamma(1.5) = 0.5\Gamma(0.5) = 0.5\sqrt{\pi}$$

So:

$$E(X) = \frac{0.5\sqrt{\pi}}{c^{\frac{1}{2}}} = \frac{0.886227}{c^{\frac{1}{2}}}$$

Also:

$$E(X^2) = \frac{\Gamma(1 + 2 \times \frac{1}{2})}{c^{2 \times \frac{1}{2}}} = \frac{\Gamma(2)}{c} = \frac{1}{c}$$

and:

$$\text{var}(X) = E(X^2) - [E(X)]^2 = \frac{1}{c} - \left(\frac{0.5\sqrt{\pi}}{c^{\frac{1}{2}}} \right)^2 = \frac{1 - 0.25\pi}{c}$$

So the standard deviation is $\sqrt{\frac{1 - 0.25\pi}{c}} = \frac{0.463251}{c^{\frac{1}{2}}}$, which is less than $E(X)$.

In fact the mean and standard deviation are equal when $\gamma = 1$.

15.4 We can calculate the value of a using the relationship between the gamma distribution and the chi-squared distribution:

$$X \sim \text{Gamma}(20, 0.1) \Leftrightarrow 2 \times 0.1X \sim \chi_{2 \times 20}^2$$

So:

$$P(X > a) = P(0.2X > 0.2a) = P(\chi_{40}^2 > 0.2a) = 0.05$$

ie $0.2a$ is the upper 5% point of χ_{40}^2 . From page 169 of the *Tables*, we see that the upper 5% point of this chi-squared distribution is 55.76. So:

$$a = \frac{55.76}{0.2} = 278.8$$

The value of a can also be determined in R using the command `qgamma(0.95, 20, 0.1)`. The R command `q` gives us the percentiles of a distribution. We follow the letter `q` with the name of the distribution. Here we want the upper 5% point, ie the 95th percentile, so the first argument is 0.95. The second and third arguments are the parameters of the gamma distribution.

15.5 (i) **Maximum likelihood estimate**

The likelihood function is:

$$\begin{aligned} L(\alpha, \lambda, \gamma) &= \prod_{i=1}^n f(x_i) \\ &= \prod_{i=1}^n \frac{\alpha \gamma \lambda^\alpha x_i^{\gamma-1}}{(\lambda + x_i^\gamma)^{\alpha+1}} \\ &= \alpha^n \gamma^n \lambda^{n\alpha} \prod_{i=1}^n \frac{x_i^{\gamma-1}}{(\lambda + x_i^\gamma)^{\alpha+1}} \end{aligned}$$

Taking logs:

$$\ln L = n \ln \alpha + n \ln \gamma + n\alpha \ln \lambda + (\gamma - 1) \sum_{i=1}^n \ln x_i - (\alpha + 1) \sum_{i=1}^n \ln(\lambda + x_i^\gamma)$$

Differentiating with respect to α :

$$\frac{\partial}{\partial \alpha} \ln L = \frac{n}{\alpha} + n \ln \lambda - \sum_{i=1}^n \ln(\lambda + x_i^\gamma)$$

This is equal to 0 when:

$$\alpha = \frac{n}{\sum_{i=1}^n \ln(\lambda + x_i^\gamma) - n \ln \lambda}$$

Since we can assume that this is a maximum, we can say that the maximum likelihood estimate of α is:

$$\hat{\alpha} = \frac{n}{\sum_{i=1}^n \ln(\lambda + x_i^\gamma) - n \ln \lambda}$$

(ii) **Numerical value**

For the sample given, we have $\sum_{i=1}^5 \ln(500 + x_i^2) = 47.6245$. So:

$$\hat{\alpha} = \frac{5}{47.6245 - 5 \ln 500} = 0.3021$$

α is the easy parameter to estimate. λ and γ are much more difficult to estimate using MLE because of the form of the last term in the log-likelihood function.

15.6 Let X denote the claim amount random variable. Then the likelihood function is:

$$L = C [P(X = 200)]^{25} [P(X = 500)]^{15} = C p^{25} (1 - p)^{15}$$

where C is a constant.

The log-likelihood function is:

$$\ln L = \ln C + 25 \ln p + 15 \ln(1 - p)$$

Differentiating this with respect to p gives:

$$\frac{d \ln L}{dp} = \frac{25}{p} - \frac{15}{1 - p}$$

Now:

$$\begin{aligned} \frac{d \ln L}{dp} = 0 &\Leftrightarrow \frac{25}{p} = \frac{15}{1 - p} \\ &\Leftrightarrow 25 - 25p = 15p \\ &\Leftrightarrow 25 = 40p \\ &\Leftrightarrow p = \frac{25}{40} = 0.625 \end{aligned}$$

So we have a stationary point when $p = 0.625$. To determine the nature of the stationary point, we check the sign of the second derivative:

$$\frac{d^2 \ln L}{dp^2} = -\frac{25}{p^2} - \frac{15}{(1 - p)^2}$$

This is negative when $p = 0.625$. (In fact, this second derivative is always negative.) So the maximum likelihood estimate of p is 0.625.

15.7 Differentiating the MGF:

$$\begin{aligned} M'(t) &= 0.4(-2)(-20)(1 - 20t)^{-3} + 0.6(-3)(-30)(1 - 30t)^{-4} \\ &= 16(1 - 20t)^{-3} + 54(1 - 30t)^{-4} \end{aligned}$$

The expected loss amount is:

$$M'(0) = 16 + 54 = 70$$

- 15.8 Let X denote the amount of a randomly chosen claim. We know that $X | \alpha \sim \text{Gamma}(\alpha, 0.8)$. So, using the conditional expectation formula:

$$E(X) = E(E(X | \alpha)) = E\left(\frac{\alpha}{0.8}\right) = \frac{1}{0.8}E(\alpha) \quad [1]$$

Then using the fact that $\alpha \sim \text{Gamma}(200, 0.5)$:

$$E(X) = \frac{1}{0.8} \times \frac{200}{0.5} = 500 \quad [1]$$

Using the conditional variance formula:

$$\text{var}(X) = E(\text{var}(X | \alpha)) + \text{var}(E(X | \alpha)) = E\left(\frac{\alpha}{0.8^2}\right) + \text{var}\left(\frac{\alpha}{0.8}\right) = \frac{1}{0.8^2}E(\alpha) + \frac{1}{0.8^2}\text{var}(\alpha) \quad [2]$$

Then using the fact that $\alpha \sim \text{Gamma}(200, 0.5)$:

$$\text{var}(X) = \frac{1}{0.8^2} \times \frac{200}{0.5} + \frac{1}{0.8^2} \times \frac{200}{0.5^2} = 1,875 \quad [1]$$

- 15.9 This is Subject 106, September 2003, Question 5.

(i) **Estimate c**

The random variable X has a Weibull distribution. Comparing the given PDF with the Weibull PDF from page 15 of the *Tables*:

$$f_1(x) = 2cx e^{-cx^2} = c\gamma x^{\gamma-1} e^{-cx^\gamma} \Rightarrow \gamma = 2$$

We are told that $P(X > 1,000)$ is expected to be 0.01 and we know that:

$$P(X > 1,000) = 1 - F(1,000) = e^{-c \times 1,000^2} \quad [1]$$

Setting this equal to 0.01 gives the estimated value of c to be:

$$\hat{c} = -\frac{\ln 0.01}{1,000^2} = 4.605 \times 10^{-6} \quad [1]$$

(ii) **Estimate λ**

Here, X has an exponential distribution and:

$$P(X > 1,000) = 1 - F(1,000) = e^{-1,000\lambda} \quad [1]$$

Setting this equal to 0.01 gives the estimated value of λ to be:

$$\hat{\lambda} = -\frac{\ln 0.01}{1,000} = 4.605 \times 10^{-3} \quad [1]$$

(iii)(a) **Calculate M**

We require M such that:

$$P(X > M) = 1 - F(M) = 0.001$$

For the Weibull distribution with $c = 4.605 \times 10^{-6}$:

$$e^{-4.605 \times 10^{-6} \times M^2} = 0.001$$

$$\Rightarrow M^2 = -\frac{\ln 0.001}{4.605 \times 10^{-6}} = 1,500,000$$

$$\Rightarrow M = 1,225 \quad [1]$$

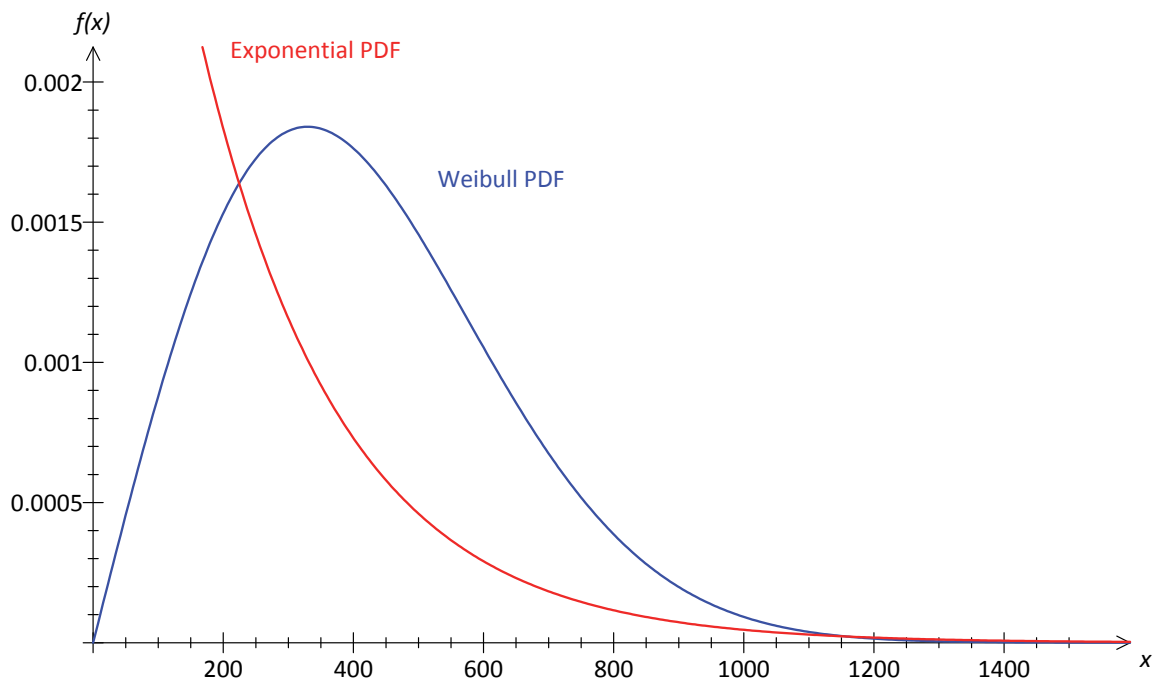
For the exponential distribution with $\lambda = 4.605 \times 10^{-3}$:

$$e^{-4.605 \times 10^{-3} \times M} = 0.001 \Rightarrow M = -\frac{\ln 0.001}{4.605 \times 10^{-3}} = 1,500 \quad [1]$$

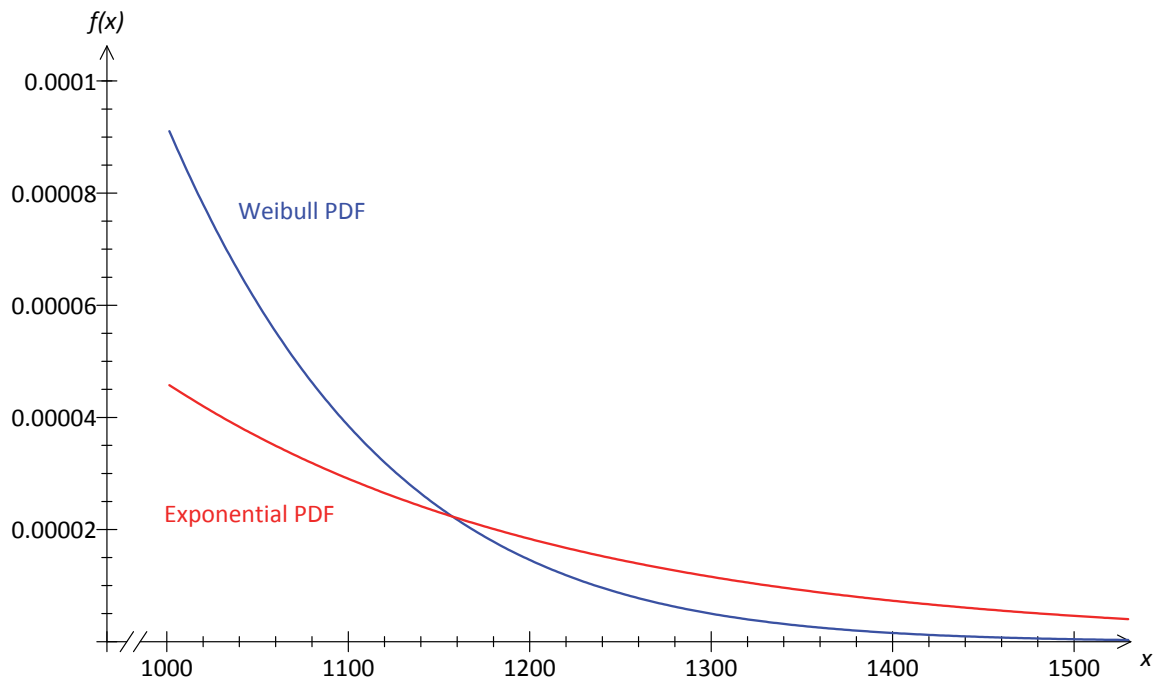
(iii)(b) **Comment**

The probability that the Weibull random variable exceeds 1,225 is 0.001 but the probability that the exponential random variable exceeds 1,225 is more than 0.001. This is because the exponential distribution has a heavier tail than the Weibull. [1]

A graph of the distributions is shown below:



Looking more closely at the tails, it is clear that the exponential distribution has a heavier tail than the Weibull distribution:



15.10 (i)(a) **Maximum likelihood estimate**

The PDF of the Weibull distribution is:

$$f(x) = c\gamma x^{\gamma-1} e^{-cx^\gamma}, \quad x > 0$$

So the likelihood function in this case is:

$$L(c) = 2cx_1 e^{-cx_1^2} \times \dots \times 2cx_n e^{-cx_n^2} = \text{constant} \times c^n e^{-c \sum x_i^2} \quad [\frac{1}{2}]$$

Taking logs, we obtain:

$$\ln L = \text{constant} + n \ln c - c \sum_{i=1}^n x_i^2 \quad [\frac{1}{2}]$$

Differentiating this with respect to c , we obtain:

$$\frac{d}{dc} \ln L = \frac{n}{c} - \sum x_i^2 \quad [\frac{1}{2}]$$

This is equal to 0 when:

$$c = \frac{n}{\sum x_i^2} \quad [\frac{1}{2}]$$

We can check that this does give us a maximum by examining the second derivative of the log-likelihood:

$$\frac{d^2}{dc^2} \ln L = -\frac{n}{c^2} \quad [1/2]$$

This is negative when $c = \frac{n}{\sum x_i^2}$. So, $\hat{c}$, the maximum likelihood estimate of c is $\frac{n}{\sum x_i^2}$. [1/2]

Substituting the sample data results into the formula from part (i)(a) gives:

$$\hat{c} = \frac{100}{976,444,000} = 1.0241 \times 10^{-7} \quad [1]$$

(ii) **Method of moments estimate**

We obtain the corresponding method of moments estimate for c by equating the sample and population means.

Using the formula for the mean of the Weibull distribution with $\gamma = 2$ (given on page 15 of the *Tables*) and the properties of the gamma function (given on page 5 of the *Tables*), we have:

$$E(X) = \frac{\Gamma(1 + \frac{1}{2})}{c^{\frac{1}{2}}} = \frac{0.5\Gamma(0.5)}{c^{\frac{1}{2}}} = \frac{0.5\sqrt{\pi}}{c^{\frac{1}{2}}} \quad [1]$$

From the data we have:

$$\bar{x} = \frac{487,926}{100} = 4,879.26 \quad [1/2]$$

Equating $E(X)$ and $\bar{x}$ gives:

$$\hat{c} = \left(\frac{0.5\sqrt{\pi}}{\bar{x}} \right)^2 = \left(\frac{0.5\sqrt{\pi}}{4,879.26} \right)^2 = 3.299 \times 10^{-8} \quad [1/2]$$

(iii) **Method of percentiles estimate**

The median of the distribution is the value of M such that $F(M) = \frac{1}{2}$.

Equating this to the sample median of 4,500 gives the method of percentiles estimate, $\tilde{c}$:

$$F(4,500) = 1 - e^{-\tilde{c} \times 4,500^2} = \frac{1}{2} \Rightarrow \tilde{c} = -\frac{\ln \frac{1}{2}}{4,500^2} = 3.423 \times 10^{-8} \quad [2]$$

15.11 Since the lower quartile is 200, we have:

$$F(200) = 0.25$$

Also, using the fact that claim amounts follow an exponential distribution:

$$F(200) = 1 - e^{-200\lambda} \quad [1/2]$$

So:

$$e^{-200\lambda} = 1 - 0.25 = 0.75 \quad [1/2]$$

Taking logs:

$$\lambda = -\frac{1}{200} \ln 0.75 = 0.0014384 \quad [1]$$

So the mean claim amount is:

$$\frac{1}{\lambda} = 695.21 \quad [1]$$

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